

FROM NOISE TO SIGNAL

Turning fragmented signals into actionable intelligence.

TEAM

CyberFusion

EVENT

IBM BoB AI Innovation Hackathon 2026

TRACK

AI + Cybersecurity + Threat Intelligence

Ingest Alerts

Continuous ingestion of fragmented telemetry, cloud alerts, and high-frequency network events.

Correlate & Score

Machine learning correlation maps multi-vector attack paths and suppresses ambient false positives.

Deliver Intelligence

Dispatches prioritized, contextualized, and enriched incident dossiers directly to SOC teams.

THOUSANDS OF ALERTS. ONE ANALYST.

The Volume Problem

SIEM systems, cyber sensors, threat intelligence feeds, and operational reports generate thousands of heterogeneous alerts daily. No analyst can process them all.

The Real Problem

Alerts arrive **fragmented, disconnected, and unpriorised**. Without context, even expert analysts cannot reliably distinguish signal from noise.

The Consequence

False positives waste critical analyst time. Genuine threats are missed. Decisions are delayed — sometimes fatally.

False Positives

Drain the hours that matter most — every minute chasing noise is a minute not spent on real threats.

Missed Threats

Fragmented signals from separate sources never converge. A coordinated attack looks like three unrelated events.

Delayed Decisions

No prioritisation means the critical alert sits in the same queue as the routine one. Response windows are lost.

Missing one genuine threat can be catastrophic. Investigating thousands of false positives wastes the time that matters most.

FROM ALERTS TO INTELLIGENCE

The CyberFusion Pipeline

1

01 — NORMALIZE

Connect related events across heterogeneous sources. One unified format, regardless of origin.

2

02 — CORRELATE

Identify patterns and relationships in fragmented signals. Surface the coordinated attack hiding in plain sight.

3

03 — SCORE RISK

Prioritise what matters most using evidence-based risk assessment. Critical threats rise to the top.

4

04 — AI INTELLIGENCE

Explain evidence, assess possible intent, generate BLUF. The analyst receives a complete intelligence picture.

Human Analyst — Final Decision-Maker

The pipeline exists to serve the analyst, not replace them. Every output is designed for rapid human review and decision.



Result: Fragmented alerts become correlated, prioritised,

ONE PIPELINE. MULTIPLE INTELLIGENCE LAYERS.

Full-Stack Architecture

Frontend: React + Vite + Tailwind

Backend: Node.js + Express

Database: MySQL

Testing: Jest + Supertest

AI Provider: Groq AI (openai/gpt-oss-120b)

01

Threat Sources

SIEM, Cyber Sensors, Threat Intelligence Feeds, Operational Reports

02

Ingestion & Normalization

Standardise heterogeneous alert formats into a unified schema

03

Persistence Layer

MySQL database for reliable state management and audit trail

04

Correlation Engine

Identify related events and patterns across source types

05

Risk Engine

Calculate authoritative risk scores and priority levels from evidence

06

AI Intelligence Layer → REST API → Human Analyst

Evidence explanation and BLUF delivered via structured API to analyst dashboard

ONE ATTACK. FOUR SIGNALS. ONE DECISION.

Verified Scenario: Multi-Source Threat Correlation

1

SIEM Port Scan Detected

Reconnaissance activity flagged by SIEM sensor

2

Failed SSH Login Attempt

Cyber sensor detects brute-force SSH access attempt

3

Failed MySQL Login Attempt

Cyber sensor detects credential attack on database

4

Threat Intelligence IOC Match

Source IP matches known malicious indicator of compromise

Intelligence Output

77

Correlation Score

VERY STRONG

81

Risk Score

CRITICAL

92%

AI Confidence

HIGH



AI Assessment: Strongly correlated multi-source threat pattern indicating coordinated reconnaissance and access attempt activity.

Recommended Action: Immediate analyst review and incident response activation.

IBM-ALIGNED AI ARCHITECTURE

How AI Adds Intelligence — Without Replacing Judgment

AI Does NOT

- Calculate authoritative risk scores
- Modify correlation results
- Invent evidence outside the supplied record
- Make final decisions on behalf of analysts

AI IS Constrained To

- Supplied evidence only — no external reasoning
- Guardrails preventing hallucination
- Out-of-boundary response detection and fallback
- Full transparency of reasoning to the analyst

AI Produces

BLUF

Bottom Line Up Front — one-line decision summary

Threat Assessment

Structured evaluation of threat pattern and severity

Possible Intent

Analysis of likely adversary objective from evidence

Evidence Summary

Concise synthesis of correlated signals and reasoning

Analyst Actions

Recommended next steps for human review and response

Provider Architecture

Provider: Groq AI (openai/gpt-oss-120b) | **Abstraction:** Flexible provider integration layer | **Human-in-the-Loop:** Analyst retains final decision authority with full transparency

FROM SIGNAL NOISE TO DECISION SIGNAL

Scenario A — Verified Output

4

Correlated Events

Across 3 source types

77

Correlation Score

VERY STRONG

81

Risk Score

CRITICAL

92%

AI Confidence

HIGH

Backend Verification — 259 Tests · 0 Failures

Test Domain	Status
REST API Health	PASS
AI Availability & Groq Provider	PASS
Persistence & Idempotency	PASS
Risk Integrity	PASS
Guardrails & Response Safety	PASS

13 test suites · 259 automated tests · 0 failures. Every critical system behaviour verified before submission.

📄 **Impact:** CyberFusion reduces analyst cognitive load by connecting fragmented signals, prioritising risk, and presenting evidence in decision-ready format — so the right decision is made faster, with confidence.

BUILT BY CYBERFUSION



[MEMBER 1]

Backend Architecture & Pipeline Engineering

Designed and built the ingestion, normalization, correlation, and risk engine layers underpinning the entire platform.



[MEMBER 3]

Frontend & User Experience

Built the analyst dashboard in React + Vite + Tailwind, translating complex intelligence output into clear decision interfaces.



[MEMBER 2]

AI Intelligence & Prompt Engineering

Architected the AI Intelligence Layer, guardrails, evidence-boundary constraints, and BLUF generation pipeline.



[MEMBER 4]

Integration, Testing & Documentation

Delivered 259 automated tests across 13 suites with zero failures, and ensured full system integration and documentation.

Project

Threat Intelligence Correlation & Alert Prioritisation Assistant

Hackathon

IBM BoB AI Innovation Hackathon 2026

Track

AI + Cybersecurity + Threat Intelligence

Team

CyberFusion — built from the ground up for a real problem with a working, verified system.

From fragmented alerts to actionable intelligence.